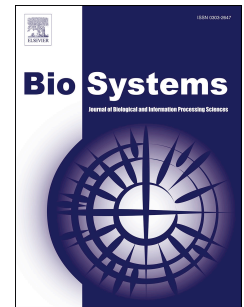


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BIOTENSEGRITY IS THE SUPER-STABILITY HYPOTHESIS FOR BIOLOGY

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BIOTENSEGRITY IS THE SUPER-STABILITY HYPOTHESIS FOR BIOLOGY

Abstract

Biotensegrity models living systems in ways that were inconceivable in the past but has taken some time to become widely accepted because of its challenges to generally accepted wisdom. Orthodox biomechanics is essentially based on mechanistic models from the seventeenth century and allowed over-simplified representations of anatomy and motion to persist to the present day, with the approximations and assumptions inherent within its methods routinely overlooked. Living organisms, however, are hugely complex, intrinsically indeterminate and exist in states that are far from equilibrium, and although their simplification within the machine model has enabled great progress in the mapping of structure to function — and benefitted our healthcare systems in remarkable ways — it has also obfuscated the foundational basis for stability, motion and life itself.

In contrast, biotensegrity is a conceptual framework that is founded on a fundamental set of self-organizing principles and includes all the complexities of life — at every heterarchical level from viruses to vertebrates and molecules to the whole organism — with stability and motion controlled from within the structure itself and the homeostatic algorithm of super-stability. Here, anatomy is no longer reduced to a set of discrete parts but becomes the physical representation of a hugely complex pattern of interacting force vectors, and which are themselves organized within a complex tensegrity configuration that enables each part of the system to adapt to its locally-changing environment in the most energy-efficient way — from embryo to adult — and remain intrinsically stable throughout.

Keywords

Algorithm; Anatomy; Biomechanics; Framework; Hierarchy; Homeostasis; Kinematics

Highlights

The biotensegrity concept is based on a fundamental set of self-organizing principles.

Tensegrity structures are the physical representations of invisible force-vectors.

Living organisms as intrinsically-stable tensegrity frameworks.

The closed-chain kinematics of tensegrity replaces the contrived machine/lever model.

Super-stability as a general mathematical framework for biological transformations.

1.0 Introduction

“The principle of organization is the fundamental source of biological stability.” (Soto et al., 2016)

Searching for the principles that underlie a particular field of enquiry has always been at the heart of scientific research, with similarities in observational data suggesting a common causality that is often formalized into a specific law or methodology (Newton, 1687; Dhar and Giuliani, 2010). Naturally, these have been taken as the standards and way-markers from which to proceed, and researchers continue to confirm their methods with examples that link the simple with the complex in a predictable way. Each step in the process follows from what preceded it and becomes established within a tradition that is reinforced by its successes (Kuhn, 1996); but it also raises a question.

Supposing the basic premise was just an assumption: a brilliant insight, well-reasoned, consistent with the data and with useful outcomes. Something that answered the questions of the day but missed or side-stepped a crucial point; what then? Such speculation might almost seem heretical except there is no suggestion that the premise was necessarily wrong. It simply highlights the fact that traditions can lead to important issues being overlooked — or pushed aside (Kuhn, 1996; Denton, 2013; Levin and Lowell de Solórzano, 2024; Scarr et al., 2024) — and which cannot be properly addressed until new concepts and methodologies enable them to be examined in ways that were not possible in the past (Bejan, 2016; Soto et al., 2016; Dornhaus et al., 2021; Mehta, 2024).

1.1 We are NOT machines

For example, the mechanistic theories of nature promulgated by Descartes (1677), Borelli (1680), and Newton (1687) are now well-established, with developments in computing, mathematics and

engineering methodologies (Freutel et al., 2014; Deriglazov, 2017) enabling great progress to be made in the modelling of living systems at every level (Roupa et al., 2022). The human body is thus widely considered as a machine with fixed interacting parts: a musculoskeletal duality of weight supporting columns, beams and arches, and muscles that change their positions through a localized lever system (Ennos, 2012; Oxland, 2016; Standring, 2020); and with a vast number of cellular ‘factories’ and molecular ‘machines’ that fabricate and provide the necessary resources for the higher-level structures to function (Piccolino, 2000; Sheetz and Yu, 2018).

The problem is that living organisms are hugely complex, intrinsically indeterminate and exist in states that are far from equilibrium (Glazier and Davids, 2009; Greenspan, 2012; Pascal and Pross, 2014), and although their simplification within the machine model has enabled great progress in the mapping of structure to function (Standring, 2020), it has also obfuscated the foundational basis for stability, motion and life itself (Dupré and Nicholson, 2018; Levin and Lowell de Solórzano, 2024; Scarr et al., 2024). Indeed, the ability to overlook the approximations, assumptions and unknowns — inherent within standard analytical methods — is routinely accepted as part of the modelling process (Freutel et al., 2014) with Russell (1931, p. 65) wryly noting that “all exact science is dominated by the idea of approximation”.

The machine metaphor has thus become embedded in orthodox thinking with the DNA/genetic code, metabolic pathways and the cell (as ‘the basic structural and functional unit of life’) now forming the basis for modern research (Pross, 2019; Rull, 2022; Moon, 2023), and most discussions about laws, principles and unifying frameworks beginning at these *relatively high levels of complexity* (Newman et al., 2006; Belousov, 2008; Dornhaus et al., 2021; Herman et al., 2021; Creath, 2023). Yet, the popular view of the cell as ‘a factory for producing molecular machines’ (Piccolino, 2000; Sheetz and Yu, 2018) says little about the underlying ‘rules of life’; and although the gene is an essential source of information in the production of proteins and other molecular processes, Nicholson (2019) noted that “...[its] products are released into a cellular milieu that *already* possesses spatial structure... shaped by pre-existing self-organizing processes” (*italics added*). So, what *are* the basic principles that enable an organism to develop, self-organize, operate in the most energy-efficient way and remain intrinsically stable — super-stable — throughout the whole of its life?

1.2 Principles, laws and concepts

The notion that there might be some special laws or principles of biology is frequently dismissed (Katsnelson et al., 2018; Lawing et al., 2021), although has its supporters (Creath, 2023; Moon,

2023), yet the complete reduction of biology to physics and chemistry is also challenging (Nicholson, 2019; Mehta, 2024). Thus, while the laws of motion and thermodynamics are considered universal, the complex processes underlying cell function, regeneration, development, reproduction and evolution — inherent within living systems — are still poorly understood in physico-chemical terms (Pross, 2019).

The field of developmental biology, for example, has long-appreciated the role of epigenetics in these processes, and that the interactions between cells and tissue masses is dependent on inherent physical laws with self-organizing capabilities (Steinberg, 1996; Newman et al., 2006; Hall and Laubichler, 2008). It also recognizes that their complex behaviours are dependent on the physics of the soft-matter materials that constitute them (Drenckhan et al., 2005; Levental et al., 2007; Levin and Lowell de Solórzano, 2024), yet there is still a need for a unifying theory.

Haldane (1931, p. 33) firmly refuted the machine model but had no doubt that “a meeting-point between biology and physical science may at some time be found” (Haldane, 1931), while Schrodinger (1944/1992, p. 80) recognized the shortcomings in traditional biophysical explanations and suggested that “we must be prepared to find a new type of physical law”, a position that still echoes today (Lawing et al., 2021; Mehta, 2024). Thus, the concept of biotensegrity as a completely different way of thinking about life — one that fully rejects the machine model and is founded on a more fundamental set of physical ‘rules’ — is now presented.

2.0 Tensegrity

The term ‘tensegrity’ is derived from the words *tension* and *integrity* and describes a structural system that consists of a set of discrete compression elements (*struts*) suspended within a network of tensioned elements (*cables*) (Figure 1A) (Sultan, 2009; Bansod et al., 2014). Thompson (2017) emphasised the importance of the tension-compression duality to an organism’s structural integrity, but it was not until 1949 that Fuller (1961; 1975) and Snelson (Snelson, 1996; Heartney, 2009) first recognized this new structural principle.

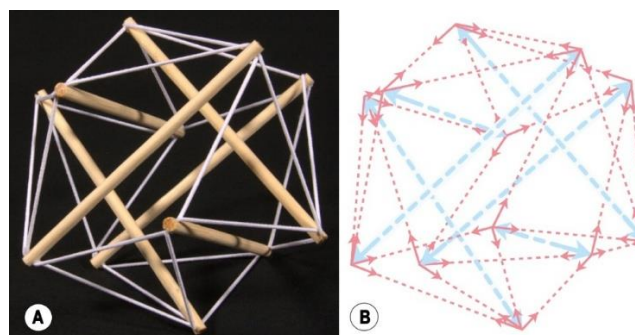


Figure 1. A) A tensegrity model (T-icosahedron/T-icosa) showing the isolated compression struts suspended within the global tension network of cables (© Rory James); **B)** the vectored diagram of physical forces (modified from Levin & Scarr, 2022 © Elsevier).

Tensegrity models are strong, relatively light in weight, flexible and resilient, and their physical properties are similar to living tissues in many ways (Ingber et al., 2014; Levin, 2015; Stamenović and Smith, 2020). They can change shape with the minimum of effort and automatically return to the same stable position of equilibrium — irrespective of the direction of gravity — and remain intrinsically stable throughout (Levin, 2014; Turvey and Fonseca, 2014). Each structural element carries its own type of load – tension *or* compression – and contributes to a pre/self-stressed structural arrangement that displays characteristic non-linear stress-strain behaviours and becomes stronger when loaded (Sultan, 2009) (Figure 2). When one part changes, *the entire system becomes involved*, thus enabling forces, power and information to be transferred from one region to another in a controlled way (Levin et al., 2017; Dischiavi et al., 2018).

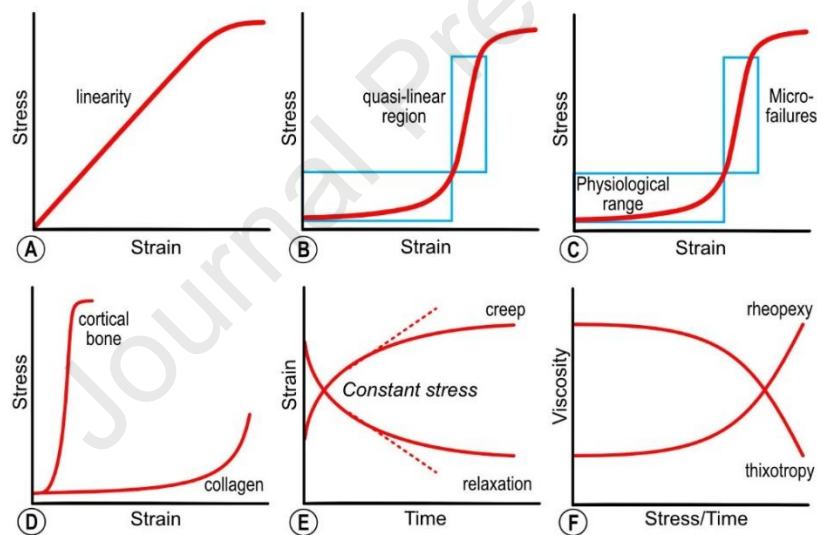


Figure 2. Typical stress-strain curves for materials that conform to **A)** Hooke's Law; **B & C)** the non-linear behaviour of tensegrity and living tissues (modified from Scarr, 2020 © Elsevier); **D - F)** some further examples of complex tissue/fluid (soft-matter) behaviours.

This non-linear stress/strain relationship, however, greatly complicates the analysis, with living tissues generally considered to function within the high stress/strain ('vertical' box) region (Figure 2B), and which is assumed to be a straight line. It is, however, much more likely that they operate within the low stress/strain ('horizontal' box) part of the curve — like tensegrity — where they can be more responsive, consume less energy and operate more efficiently: all defining characteristics of living systems (Levin, 2009) (Figure 2C); and as Harris (2018) noted: "to regard Young's Modulus as a

constant... is a very big simplifying assumption” because it will be different at each point along the non-linear curve. The intrinsic tension contained in both tensegrity models and living tissues also means that the initial part of their stress-strain curves never starts at zero stress, and that they are always ‘primed for action’ (Masi and Hannon, 2009) (Figure 2B).

2.1 Applications

Tensegrity has found a place in constructivist art (Snelson, 1996; Gough, 1998; Heartney, 2009), architecture (Gomez-Jauregui et al., 2023), the development of novel types of materials (Al Sabouni-Zawadzka and Gilewski, 2018; Liu et al., 2019; Bauer et al., 2021), robots destined for the exploration of space (Sunspiral et al., 2013; Caluwaerts et al., 2014; Rimoli, 2016), new classes of joint and prostheses with potential medical benefits (Jung et al., 2021; Brandão and Savi, 2022; Oh et al., 2022; Dong et al., 2025), and biology in general (Levin, 2015; Scarr, 2018; Lowell de Solórzano, 2020).

It has been described in relation to molecules (Edwards et al., 2012; Bywater, 2017; Reilly and Ingber, 2018; Azulay et al., 2022), viruses (Sitharam and Agbandje-Mckenna, 2006), cells (Ingber et al., 2014; Fraldi et al., 2019), soft tissues (Boghdady et al., 2021; Ingber, 2023; Scarr et al., 2024) and regional anatomy, including the cranial vault (Scarr, 2008; Looman et al., 2022), temporo-mandibular joint (Scarr and Harrison, 2017), spine (Levin, 2002), shoulder (Levin, 1997), elbow (Scarr, 2012), pelvis (Levin, 2007; Pardehshenas et al., 2014), knee (Kim and Turvey, 2018), foot (Wilson and Kiely, 2016), heart (Parker and Ingber, 2007), lung (Ingber, 2023), brain (Van Essen, 2023), postural control (Cabe, 2019; Mangalam et al., 2020), homeostasis (Stamenović and Smith, 2020), running (Kiely and Collins, 2016; Wilson and Kiely, 2016) and the body as a whole (Levin, 2015; Dischiavi et al., 2018; Scarr, 2018; Levin and Lowell de Solórzano, 2024).

Tensegrity is also appreciated within cancer research (Dinicola et al., 2011; Tadeo et al., 2014), sports science (Kiely and Collins, 2016; Roy et al., 2025), ecological psychology (Turvey and Fonseca, 2014; Cabe, 2019), Constructal Law (Scarr, 2018; Rehnke, 2024), soft condensed matter (Levin and Lowell de Solórzano, 2024), processual biology (Dupré and Nicholson, 2018), and clinical reasoning (Fullerton, 2018; Crowle and Harley, 2020; Scarr, 2020), illustrating the potential of this relatively new approach.

However, before proceeding, it should be noted that the *engineering* applications are based on well-established traditions and from which Skelton et al. (2001) introduced a tensegrity classification that does NOT apply to biology (Scarr, 2025a). So, while the development of engineered structures depends on the particular design, material properties and practicality of construction etc., living

systems have refined *themselves* over hundreds of millions of years of evolution and are constrained only by some basic principles of self-organization. It is thus likely that they have already found the most energy-efficient ways of doing things, and as a consequence, Levin (2002) introduced the term *biotensegrity* in order to make a clear distinction between living and engineering tensegrity.

Of course, both fields are closely linked and can benefit from each other (Scarr, 2025a; Zhang et al., 2025), but while the term *tensegrity* applies to the structural principle, *biotensegrity* relates specifically to living systems; and although much of the bio-focussed literature has used the word *tensegrity* out of tradition (Ingber et al., 2014; Turvey and Fonseca, 2014; Boghdady et al., 2021), it should be noted that these terms are *not necessarily* synonymous.

3.0 Biotensegrity

3.1 Principles of self-organization

“At a most basic level, atoms interact with each other in a number of different ways — van der Waal’s forces, ionic bonding, hydrogen bonding, covalency and steric-repulsion etc. — all of which can be categorized into those that ‘pull’ atoms together and others that keep their centres apart. Of course, there is much more detail associated with these processes, but simplifying then into this duality of attraction and repulsion reveals them to be the drivers that cause atoms to spontaneously organize into the most stable and energy-efficient configurations, as ‘rigid’ crystals and more flexible molecules (Reilly and Ingber, 2018; Eichenauer and Lordick, 2019; Van Horn and Wang, 2021) (Figure 3).” (Scarr, 2025a).

“Such arrangements form... because of some fundamental principles of self-organization: geodesic geometry, close-packing and the minimization of energy — and which apply at *every* level — where *attraction and repulsion* at higher size-scales are now referred to as *tension and compression* and can be modelled through tensegrity (Fuller, 1961; Levin, 1982). Crystals and molecules are then close-packed tensegrity configurations of atoms with invisible struts (Edwards et al., 2012; Bywater, 2017; Reilly and Ingber, 2018; Eichenauer and Lordick, 2019).” (Scarr, 2025a)

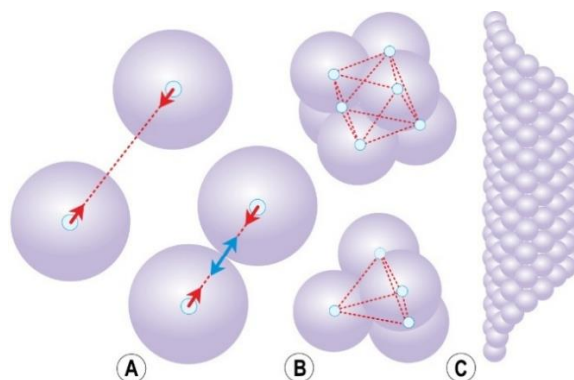


Figure 3. Schematic showing **A)** the attraction and repulsion between atoms, and some close-packing arrangements within **B)** crystals and **C)** molecules.

“Tensegrity systems are inherently geodesics because their structural members transfer tension and compressional forces along the shortest pathways and in straight lines. They are close-packed because their component parts *automatically* get as close as they can to each other, and they *always* settle into a state of minimal-energy. The tension elements pull the nodes together — and the compression elements keep them apart — with the tensegrity framework being the embodiment of these three basic principles and the *physical representation of the invisible forces within it* (Fuller, 1961; Levin, 1982; Heartney, 2009) (Figure 1).” (Scarr, 2025a)

So, rather than anatomy being represented as a collection of ‘bits’ that operate in a local piecemeal-like way (like an engineered machine), a biotensegrity perspective now models the hidden force vectors within through a unified system of closed-chain kinematics (Figure 1B).

3.2 The mechanics of life

“Closed kinematic chains (CKCs), or linkage systems, have long been employed in engineering because they provide a simple and efficient mechanism with multiple applications. Here, multiple *bars* are coupled into continuous mechanical loops with each one influencing the position, motion and stability of all the others in the system and enabling the controlled transfer and amplification (or attenuation) of force, speed and kinetic energy (Figure 4).” (Scarr, 2025a)

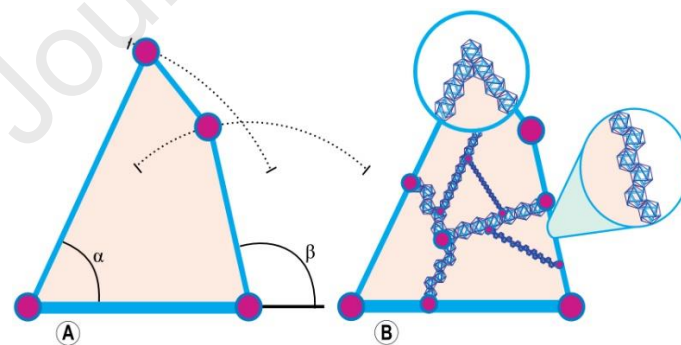


Figure 4. **A)** A 4-bar mechanism showing the ‘pin-joints’, trajectories and limits of three moving bars in relation to a fixed (horizontal) reference bar, and the changing angles (α/β) that define its non-linear behaviour (modified from Levin et al., 2017 © Elsevier); **B)** coupled linkages within a complex tensegrity hierarchy (modified from Scarr, 2025b © Handspring).

CKCs have also been described in the feeding mechanisms of fish (Müller, 1996; Farina et al., 2019), shrimps (Patek, 2019) and humans (Scarr and Harrison, 2017); the fascial connective tissues (Levin et

al., 2017) and the limbs of horses and humans (Valero-Cuevas et al., 2007; Levin et al., 2017; Olsen, 2019; Burgess, 2021).

Kinematics is all about the geometry of motion and conventionally described through algebraic equations, yet this graphic portrayal reveals things that might otherwise not be appreciated (Müller, 1996), and which is particularly relevant here because closed-chain kinematics constitutes the basic mechanics of tensegrity (Levin et al., 2017) (Figure 5). The 4-bar mechanism is thus highlighted because it enables the structure itself to control the position and motion of every part of the system (Valero-Cuevas et al., 2007) — from molecules to the complete organism — with the 3-bars creating the stability that allows the 4-bars to change shape, while those with five or more bars would be uncontrollable on their own but can be part of more complex systems (Rimoli, 2016).

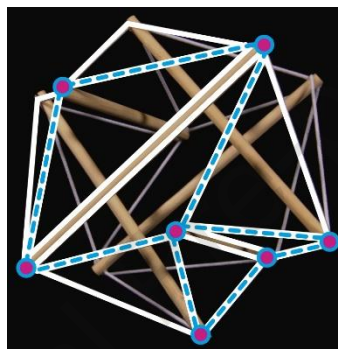


Figure 5. The T-icosa model (© Rory James) highlighting some of the fixed 3-bars (white) and flexible 4-bars (blue dotted) that guide changes in shape (reproduced from Scarr, 2018 © Handspring).

Here, each anatomical ‘bar’ (cable *or* strut) and ‘node/vertex/pin-joint’ consists of multiple sub-structures organized in a similar way (Figure 4B), with each one under tension *or* compression and mechanically integrated with all the others — at multiple levels — so that when one part changes in length or position, the entire system becomes involved. Mono-articular muscles might then be considered as variable-length bars within 3-bar systems, while bi-articular isometric muscles can form tensionally stiff bars within 4-bar systems that enable multiple joints to be coupled in ways that transfer power from one part of the body to another in a controlled way (Levin et al., 2017; Dischiavi et al., 2018) (Figure 6).

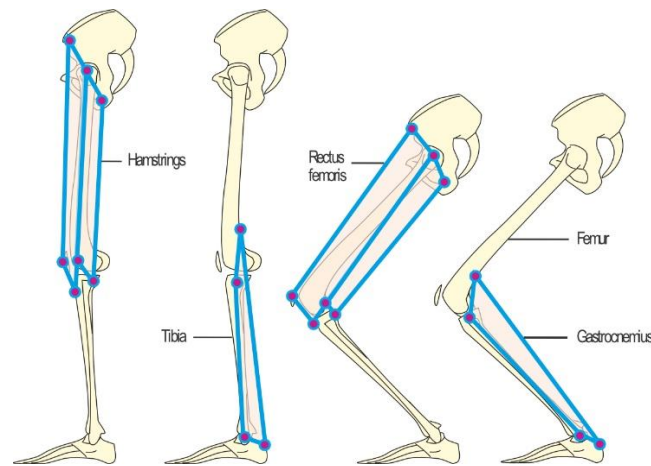


Figure 6. Schematic showing some of the (high-level) closed-chain geometries that guide changes in position of the bones in relation to each other. Reproduced from Scarr, 2025b © Handspring

Of course, such descriptions are gross simplifications of the more complex reality, but do provide an explanation for the non-linear behaviour of living systems (Müller, 1996) (Figure 2B) and show how movement can be controlled in ways that are beyond the sole capability of the nervous system (Profeta et al., 2020; Scarr et al., 2024). Here, the tissues (linkages) would be able to respond *instantly* to rapidly changing physical forces and avoid the potentially damaging consequences of transient muscular latencies (Lin et al., 2019; Del Vecchio, 2023; Scarr et al., 2024), with muscle-generated forces transmitted over a wider field than is generally appreciated (Levin et al., 2017; Dischiavi et al., 2018) and functional degeneracy built into the complexity of the system (Greenspan, 2012; Kiely and Collins, 2016; Wilson and Kiely, 2016).

The planar 4-bar is thus highlighted because of its mechanical simplicity — and as the first iteration of the *omni-directional* CKC/tensegrity heterarchy — with the system as a whole controlling the flow of kinetic energy, and with evolutionary consequences (Müller, 1996; Bejan, 2016; Kiely and Collins, 2016; Muñoz, 2019). Closed-chain kinematics and tensegrity thus present a complete whole-body biomechanical system that replaces the machine/lever orthodoxy and greatly expands the potential for biological modelling at every level (Scarr, 2025a).

3.3 Structure, stability and energetics in context.

To reiterate, the concept of biotensegrity is predicated on anatomy as *the physical representation of a hugely complex pattern of interacting force vectors* (Figure 1B), and that it is the efficient separation of these into different structural modules ('cables/struts/bars') — and their closed-chain interactions (Figure 4 & 5) — that allows such systems to change shape whilst remaining intrinsically stable throughout. However, this now raises an issue because *stability* in dynamic systems is

inseparable from its opposite: *change*, and as far-from-equilibrium systems are dependent on an external source of energy, they are inherently *unstable* in thermodynamic terms.

3.3.1 Kinetic barriers

Pross and Khodorkovsky (2004) reviewed the widespread assumptions of thermodynamic theory and showed that stability in dynamic systems relates more to their kinetic status, rather than the energetic consequences of the Second Law, where a system can be prevented from progressing towards a more thermodynamically stable state because of kinetic barriers (Pross, 2019). For example, the strong covalent bonding of carbon atoms is crucial to maintaining the stability of complex macromolecules, but such bonds are also vulnerable to the activity of specific enzymes. Under certain conditions, these enzymes can overcome the high kinetic barriers — and deep energy wells associated with covalency — by disrupting the bonds and causing the macromolecule to dissociate, thus allowing the system to progress *towards* a thermodynamically more stable/lower-energy state. In other words, depending on the particular conditions, the kinetic barriers can prevent the enzymes from reacting so that the macromolecule remains *stable* in a temporal (low entropy) sense but *unstable* in an energetic/thermodynamic (high entropy) sense because of this vulnerability (Pascal, 2025).

Pascal and Pross (2014; 2015) also revealed the mathematical basis for a group of persistently stable systems that rests solely on the kinetic power of exponential growth, a type of stability associated with replicative entities that form steady-state populations and maintain themselves over time: temporal stability through self-replication, or *dynamic kinetic stability* (DKS) (Pross, 2019). Here, such systems are characterized by the ability to catalyse their own reactions and evolve into dynamically more complex forms (Plasson et al., 2011), and which are quite different from the *static* kinetic stability typical of most inorganic chemistry; and biological metabolism can also be considered as a vast autocatalytic population of interacting molecules that reproduces itself and automatically becomes more complex (Pross and Pascal, 2023).

It is, of course, the steady-state *population* that is temporally stable rather than the individual entities within it, and its survival depends on kinetic barriers and the continual supply of resources. (Pascal and Pross, 2014; Bejan, 2016; Pross, 2019). The inherent increase in complexity associated with autocatalysis then carries the potential for different parts of the system to deal with those resources and metabolic products in different ways, thus enabling new sub-populations with new functions to emerge (Plasson et al., 2011). Here, the fundamental drive towards greater *energetic stability* enables the kinetically more stable regions to dynamically displace those that are less

stable, and the system as a whole to evolve into an even more complex and stable steady-state (Pascal and Pross, 2015). Stability, function and complexity are thus coupled in an intertwined relationship that applies to the processual self-organizing dynamics of molecular and cellular populations (Dupré and Nicholson, 2018) and underpins the far-from-equilibrium dynamics of embryological development and life in general (Kurakin, 2009; Pascal and Pross, 2014; Toman and Flegr, 2017; Pascal, 2025).

3.3.2 Constructal Law

The continuous flow of resources is thus a characteristic feature of living systems, with the ability to transform themselves in ways that enable those resources to flow more easily described through the Constructal Law (Bejan and Lorente, 2013; Bejan, 2016). Natural flow systems will *always* morph into configurations that provide greater access to the currents that flow through them, and the fractal-like patterns of rivers, trees, vascular and respiratory systems are easy examples. They are architectural solutions to increasing flow within a constantly evolving system, with these particular morphologies forming in response to the driving currents and local constraints in the changing environments through which they flow. Such morphing flow systems thus apply to every aspect of a developing and self-sustaining organism: from molecules that diffuse through a fluid environment and interact with each other in complex ways (Kurakin, 2009), to the flow of fluids through tubular systems, and structural entities that carry the forces of tension and compression (tensegrity).

So, as an organism's structural framework is constantly changing, there must be some general homeostatic algorithm that keeps all its internal processes orderly, but until recently, there has been no ready-made math for determining whether such a framework really is intrinsically stable, which is why biomechanics has been dominated by heuristic approaches that can at least provide partial solutions (Ennos, 2012; Freutel et al., 2014). Connelly and Back (1998), however, examined the spider web through the mathematics of tensegrity and highlighted the algorithmic framework of super-stability (Connelly and Guest, 2022): a fundamental theorem that now formalizes biotensegrity as a viable model for living systems (Zhang et al., 2013).

3.4 Super-stability: the ultimate abstraction

A general mathematical framework is the highest level of abstract reasoning that encompasses a particular set of variables, such as the duality of tension and compression, and is part of a logical thought process that is particularly applicable to tensegrity because of its inherent purity and simplicity (Sultan, 2009; Zhang et al., 2013; Yang et al., 2019; Connelly and Guest, 2022; Wang et al., 2023). Here, the framework can be represented through graph theory as a set of points (*nodes*,

vertices) connected by geodesic lines (*edges*) with simple distance constraints (Koutrouli et al., 2020), with certain point combinations able to get closer together but not further apart (cf. tension elements), and others that can get further apart but not closer together (cf. compression elements). In other words, the connecting edges behave like tension and compression springs that can change in length and vary the amount of energy stored inside each one (Figure 7).

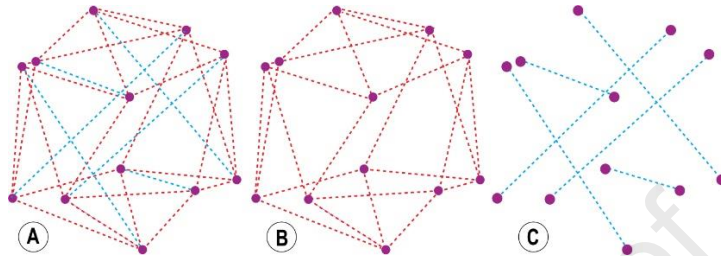


Figure 7. A) Tensegrity portrayed as an ‘energy’ framework (Figure 1B); **B)** the isolated set of points (nodes, *vertices*); **C)** point combinations (*edges*) that can get closer together but not further apart; **D)** point combinations (*edges*) that can get further apart but not closer together (modified from Scarr, 2025a © Springer).

“The stability of the framework is then achieved through “coupled changes in the elastic strain energies as the springs [edges] automatically seek their minimal-energy states, and because these energy functions are split between positive (tension) and negative (compression), such a system will *always* resolve itself into the most stable, balanced and energy-efficient configuration at any given *moment* in time; but as a *super-stable* framework, it can have ANY number of possible edge combinations for ANY number of vertices and without restricting the assignments”. (Scarr, 2025a)

“Here, the edges can be freely split into shorter ones with new vertices, thereby *increasing* their total count within the framework, or connected into longer ones that *decrease* the number of vertices, and all without compromising the system’s stability. In other words, the edges can get longer, shorter, divide, merge together or even switch energy functions with the entire system involved in making these transitions, and the configuration as a whole remaining completely stable, *no matter how complicated it becomes*”. (Scarr, 2025a)

Full mathematical definitions, derivations and theorems are given elsewhere (Connelly and Back, 1998; Sultan, 2009; Yang et al., 2019; Connelly and Guest, 2022) but the value of this as a representation of multi-scale anatomy and dynamics has already been recognized (Zhang et al., 2013) and cannot be overstated.

“At the highest, most abstract level, super-stable tensegrity provides the intrinsic-stability solution for ANY comparable framework of edges and vertices, regardless of the number of elements within

it, and it does so through the *minimal-energy principle*. Super-stability is essentially a representation of infinity, and the imagery of such a framework that has the capacity to freely switch between an *infinite* number of different combinations of edges and vertices, then allows a shift to be made from the virtual, abstract and logical perspective to the more tangible morphological transformations of biology and its processes. Here the local, regional and global meta-stabilities of attraction-repulsion and tension-compression are constantly re-organizing *themselves* in response to the ever-changing environment, and enabling each 'part' to change shape, grow, develop, transform, remodel, repair and move with the minimum of effort with the system [as a whole] remaining intrinsically stable throughout". (Scarr, 2025a)

In summary, an appreciation of living organisms as complex, constantly-changing, multi-scale energy frameworks that continuously flow from one stable state to another and maintain overall viability through their myriad homeostatic processes (Dupré and Nicholson, 2018; Stamenović and Smith, 2020).

4.0 Discussion

The notion that there might be some unifying principles governing the form and behaviour of living systems has been around for millennia, yet there is still no general agreement about their identities (Denton, 2013; Lawing et al., 2021; Creath, 2023). For the past two-hundred years, the debate over the fundamental nature of organic forms has been between the *structuralist* view based on immutable biophysical laws, and more *functionalist* theories that relate to the purposes those forms serve, and it is the latter that currently dominate (Mayr, 2004; Denton, 2013).

Up until the mid-nineteenth century, structuralism considered biological shapes, patterns and designs to be the result of particular combinations of internal processes, or lawful constraints inherent within the system (Owen, 1849, p. 2; Amundson, 2007), and in a way analogous with the formation of crystals (Schwann, 1847, p. 212). It was then inevitable that the failure to define such laws would lead to the inclusion of more functionalist ideas, such as those of Lamarck (1809/1963, p. 106), where living systems were able to adapt in response to environmental conditions: a secondary phenomenon external to the organism.

The general acceptance of Darwin's theory of evolution (and later variants) (Layer, 2022; Corning, 2023; Bojinov, 2024) then sealed the functionalist paradigm in biological thinking, where organic forms were no longer the product of natural law but contingent mutable assemblages put together piece-by-piece over time (Denton, 2013). Small, inheritable adaptations that increased an individual's ability to compete, survive and reproduce, and accumulated in its descendants in ways

that ultimately led to new species and different types of organism, i.e., an assessment of living systems as contingent collections of anatomical ‘parts’ that served some useful purpose and conveniently fitted with the machine analogy passed down from the seventeenth century (Darwin, 1862, p. 283-284)

4.1 Inherent ‘laws’ of biology

The Cartesian model has thus dominated because of its analytic convenience in virtually every field of research, but largely ignored the underlying principles that guide the processes of normal development and homeostasis (Freutel et al., 2014), with exceptions (Thompson, 1917; Newman et al., 2006; Belousov, 2008; Kurakin, 2009; Denton, 2013; Bejan, 2016; England, 2022). Fuller (1961), however, had long been looking for practical engineering solutions in the natural world because of its structural and functional efficiency, but rejected the machine model in favour of a theory based on the study of nature’s coordinate system (Fuller, 1975; Edmondson, 2007; Eichenauer and Lordick, 2019). He perceived that all natural forms were the result of matter being acted upon by hidden physical forces and proposed them as finite energy systems consisting of attraction and repulsion (tension-compression) (Figure 3), and from which the structural principle of *tensegrity* then emerged (Fuller, 1961; Snelson, 1996). So, although Fuller’s writings can seem rather cryptic and are widely dismissed, it is his ideas that ultimately led to an appreciation of this particular set of self-organizing principles and the integrated biotensegrity concept in biology (Levin, 2015; Scarr, 2018; Lowell de Solórzano, 2020; Ingber, 2023).

4.2 Stability before mobility

Biotensegrity is thus relatively new but taken some time to become widely accepted because of its challenges to generally accepted wisdom. Orthodox biomechanics has tended to focus on mobility (with stability as a secondary consideration) and employed a lever model, based on the assumption that the human body is intrinsically *unstable* and requires an external controller — the neural system — to provide the necessary ‘catch-up’ stability for transient muscular latencies (Lin et al., 2019; Del Vecchio, 2023), yet it is likely that the vulnerabilities inherent within such a mechanism would severely compromise living tissues (Scarr et al., 2024).

Scarr et al. (2024) examined the anatomy of the human musculoskeletal system from historical and more recent perspectives, and revived a long-forgotten aspect of connective tissue anatomy that was well-recognized in the seventeenth and eighteenth centuries (Malpighi, 1661; von Haller, 1747), but gradually faded in significance as Schwann’s (1847) cell theory and the machine model assumed functional dominance: the fascial system as “a *tensioned* fibrous network encompassing a complex

heterarchy of regionally specialized [close-packed] compartments under *compression*, each of which has its own physical and parenchyma-driven characteristics that contribute to the functional whole” (Scarr et al., 2024). In other words, a dynamic but *intrinsically-stable* architectural framework that includes every aspect of an organism’s anatomy and in a substantially different way from traditional models — one with its processes controlled through the mechanics/energetics of closed-chain kinematics, dynamic kinetic stability and the homeostatic algorithm of super-stability — and addresses the question of how biotensegrity relates to the future of biology and medical research (Kiely and Collins, 2016; Scarr, 2020).

4.3 Resolving the empirical dilemma

Modern technology has provided the tools to analyze living systems at every level, with mathematics — the ultimate in reductionism — successfully employed in every aspect of scientific research because it has no inherent biases (Freutel et al., 2014; Dornhaus et al., 2021; Roupá et al., 2022). It is simply a way of finding solutions and answers through the application of basic principles and rational logic but is *totally dependent on the quality of information it has to work with*, which is why the machine model and associated engineering methodologies have been brought into question (Sec. 1.1).

Biotensegrity is a *complete* whole-body concept, yet living systems are intrinsically indeterminate and probabilistic at every level (Greenspan, 2012; Pasca and Pross, 2014), and if developmental biology, evolution and biomechanics (etc.) are to be empirically addressed in this context, then new approaches that fully appreciate the significance of this will be required (Sec. 2.1) (Reilly and Ingber, 2018; Koutrouli et al., 2020; Mangalam et al., 2020; Jung et al., 2021; Azulay et al., 2022; Looman et al., 2022; Oh et al., 2022; Alicea et al., 2023).

5.0 Conclusion

Paradigms are theories that have become so embedded in our thinking that they are almost beyond question, yet challenging their assumptions is a normal part of the scientific process (Kuhn, 1996). The Cartesian tradition in biological research has thus persisted because of its successes, convenience and the lack of viable alternatives (Mayr, 2004), but its inadequacies have long been questioned (Haldane, 1931; Russell, 1931; Levin, 1981; Schrodinger, 1992; Denton, 2013; Scarr et al., 2024). Fuller’s (1961; 1975) approach to biology was then substantially different because he considered natural forms to be finite energy systems consisting of a complex set of hidden force vectors (Figure 1B & 3) and recognized that their geometric relationships within a tensegrity configuration was a key to understanding their complex behaviours (Figure 4,5).

The dissatisfaction with traditional biomechanical models and an appreciation of the self-organizing principles of geodesic geometry, close-packing and minimal-energy — inherent within all natural systems (Fuller, 1961; Levin, 1982) — then ultimately gave rise to the concept of biotensegrity (Levin, 2002) with stability and motion controlled from within the structure itself and in synergy with the nervous system (where befits): a complex, constantly changing, super-stable framework of interconnected processes that maintains the viability of the system (Dupré and Nicholson, 2018; Pross, 2019) and provides a more complete model of biology.

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Conflict of Interest Statement

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Declaration of Interest Statement

The authors declare that there are no conflicts of interest.